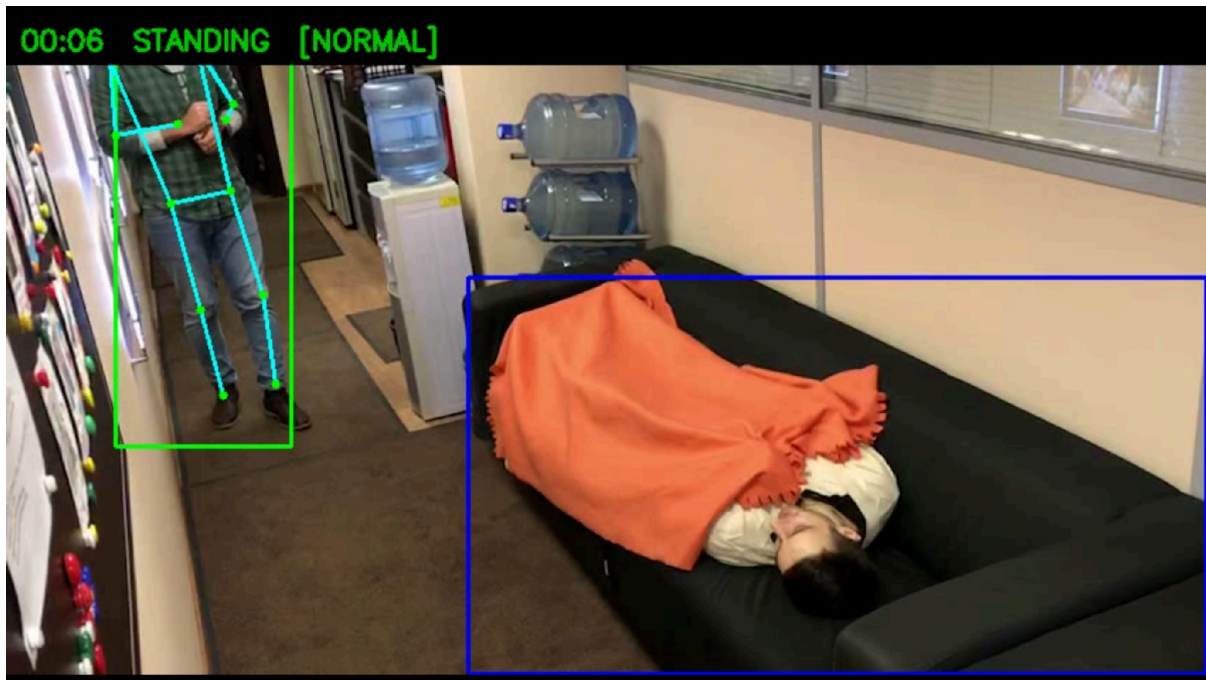


Failure Case 1: Caregiver taken as the patient



Output vs. truth: The system reports STANDING [NORMAL], but the patient is lying on the bed under a blanket.

Cause: The blanket hid the patient, so YOLO never detected them as a person. The only person detected was the caregiver, so the system picked the caregiver as the patient and reported their state instead.

Fix: Only accept someone as the patient once they are seen lying or sitting on the bed; if nobody qualifies, mark the patient as UNKNOWN and ask the VLM whether someone is under the blanket.

Failure Case 2: Fall in front of the bed read as lying in bed

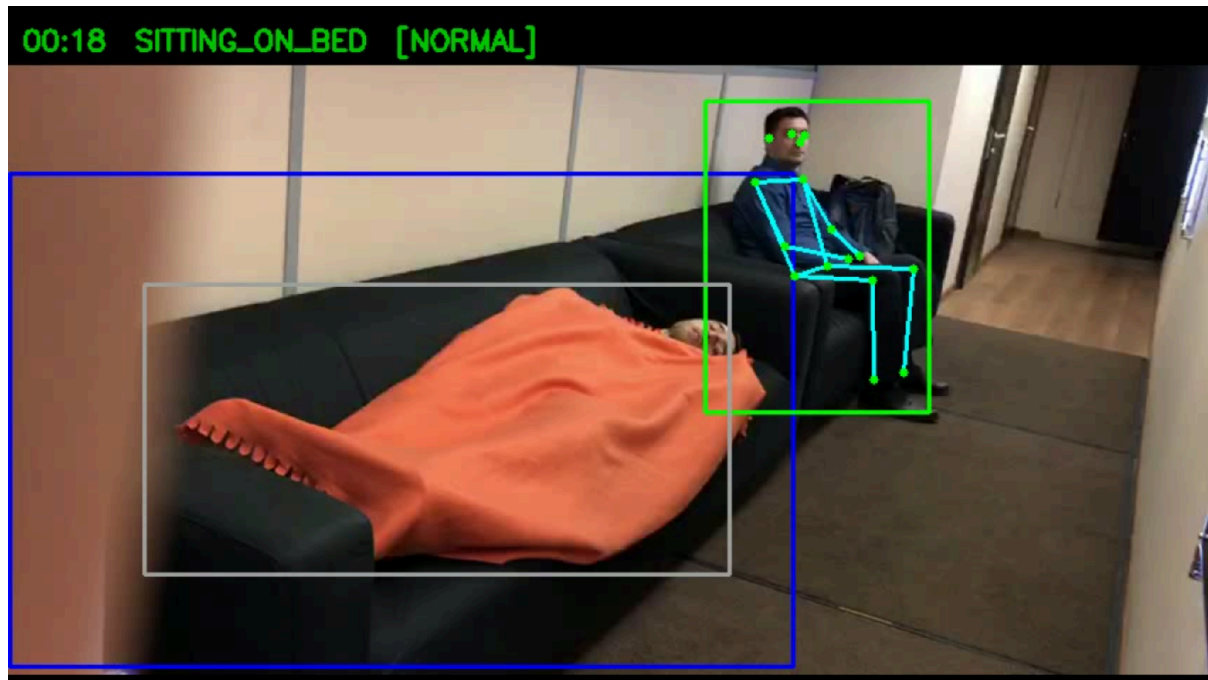


Output vs. truth: The system reports LYING_IN_BED [NORMAL], but the person is lying on the floor in front of the bed, a possible fall that should be an ALERT.

Cause: The bed region is a 2D box, and from this low camera angle it also covers the floor in front of the bed. The person's hips fall inside that box, so the rules treat them as being on the bed.

Fix: Mark the mattress surface with a hand-drawn polygon instead of the detected box, and treat lying with the hips near or below the bed's lower edge as outside the bed, confirmed by the VLM.

Failure Case 3: Wrong person kept as the patient



Output vs. truth: The system reports SITTING_ON_BED [NORMAL] for the visitor sitting at the end of the sofa, while the real patient is lying under the blanket (gray box, detected but not followed).

Cause: Once a track is chosen as the patient, the choice is never revisited. The visitor sat inside the bed region, so his hips counted as being in bed and he won the patient score; when the covered patient was detected later, the system kept following the visitor.

Fix: Weight lying in bed far above sitting when scoring the patient, re-check the patient choice over time, and ask the VLM which person is lying in the bed when two people are in the bed region.